

Three-Level Audit of Normative-Frame Sensitivity in LLM-Assisted Policy Simulation

Model-level

H1: Composite orientation

$$\text{PVOC}_m \rightarrow \bar{\Delta}_m$$

Does aggregate policy-value orientation predict cross-model burden effects?

N = 15 model endpoints

$$\beta_1 = -2.14, 95\% \text{ CI } [-6.23, 1.95]$$

$$R^2 = 0.09$$

No aggregate signal

Narrative-level

H2: Narrative frame association

$$\text{Frame}_{mir} \rightarrow \Delta_{mir}$$

Do naturally expressed rationale frames track simulated burden effects?

Model $\times$ profile pair
fixed effects

N = 2,019

Access barriers: $b = 0.13$, n.s.

Collective responsibility: $b = 0.15$, n.s.

Coercive backlash: $b = 1.21$, n.s.

No robust within-pair signal

Within-model

H3: Frame manipulation

$$\text{Assigned Frame}_c \rightarrow \Delta_{mcir}$$

Does an assigned prompt-frame clause shift the simulated burden effect within the same model?

4 models $\times$ 27 profiles
 $\times$ 4 frames

Autonomy: +4.96

Equity/access: +4.84

Collective obligation: -10.88

Frame-sensitive

Model-specific heterogeneity

Aggregate orientation: no signal · Natural rationale frames: no robust within-pair signal

Assigned framing: shifts simulated effects